

GLOBAL SALES AND FINANCIAL PERFORMANCE ANALYSIS USING POWERBI

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INTRODUCTION:

In today's business environment, organizations generate large volumes of sales and financial data through different products, customers, and sales channels. Analysing this data helps businesses understand their sales performance, costs, profitability, and customer demand. Effective analysis of such information can help identify important trends and provide a clear understanding of overall financial performance.

Microsoft Power BI is a business intelligence and data visualization tool that converts raw business data into interactive and meaningful reports. In this project, Power BI is used to analyse global sales and financial performance based on key measures such as total quantity, sales amount, total cost, profit, and profit margin. The dashboard uses different charts, KPI cards, and interactive filters to present the data in a clear and accessible manner.

The main focus of this project is to develop an Interactive Global Sales and Financial Performance Dashboard. The dashboard provides insights into sales across different channels, product performance, order priorities, and profitability. By combining financial measures with interactive visualizations, the project demonstrates how Power BI can be used to analyse business performance and support data-driven decision-making.

OBJECTIVES:

- To analyse global sales and financial performance using Microsoft Power BI.
- To evaluate key performance indicators such as Total Quantity, Total Sales, Total Cost, and Profit Margin.
- To analyse sales performance across different sales channels such as Partner, Retail, Online, and Phone.
- To compare profitability across different order priorities.
- To analyse sales performance across different products using Product Key.
- To identify patterns and variations in sales, cost, quantity, and profit.
- To create an interactive dashboard using charts, KPI cards, and slicers for effective data analysis.
- To provide meaningful business insights that can support data-driven financial and sales decisions.

DATASET DESCRIPTION:

The dataset was imported into **Microsoft Power BI** for data preparation, analysis, and visualization. The available fields were used to create key performance indicators and interactive visualizations. The dashboard focuses on understanding sales

distribution across different channels, product-level sales, and profit according to order priority.

Key Fields in the Dataset

- **Product Key** – Unique identifier used to distinguish different products.
- **Quantity** – Represents the number of units sold.
- **Sales Amount** – Indicates the total revenue generated from sales.
- **Cost** – Represents the cost associated with the products sold.
- **Profit** – Shows the amount earned after considering the relevant costs.
- **Unit Price** – Represents the selling price of an individual unit.
- **Channel** – Identifies the sales channel, such as Partner, Retail, Online, and Phone.
- **Order Priority** – Indicates the priority level of an order, such as Low, Medium, and High.

DATASET PREPARATION:

Before developing the dashboard, the dataset was prepared in **Power BI** to ensure that the data was suitable for analysis and visualization. The data was reviewed and organized so that

the required sales and financial fields could be used effectively in calculations and visualizations.

- **Imported the dataset into Power BI Desktop** for analysis and dashboard development.
- **Checked the data types** of numerical fields such as Quantity, Sales Amount, Cost, Profit, and Unit Price.
- **Verified categorical fields** such as Channel and Order Priority for accurate grouping and comparison.
- **Reviewed the Product Key field** to ensure that products could be analysed individually.
- **Checked the data for consistency** before creating charts and KPI cards.
- **Prepared the required measures** for Total Quantity, Total Sales, Total Cost, and Profit Margin.
- **Organized the fields** according to their purpose to create meaningful visualizations and interactive slicers.

DAX MEASURES USED:

DAX (Data Analysis Expressions) measures were created in Power BI to calculate the key financial and sales indicators required for the dashboard. These measures provide summarized values that dynamically respond to the selected filters and visuals.

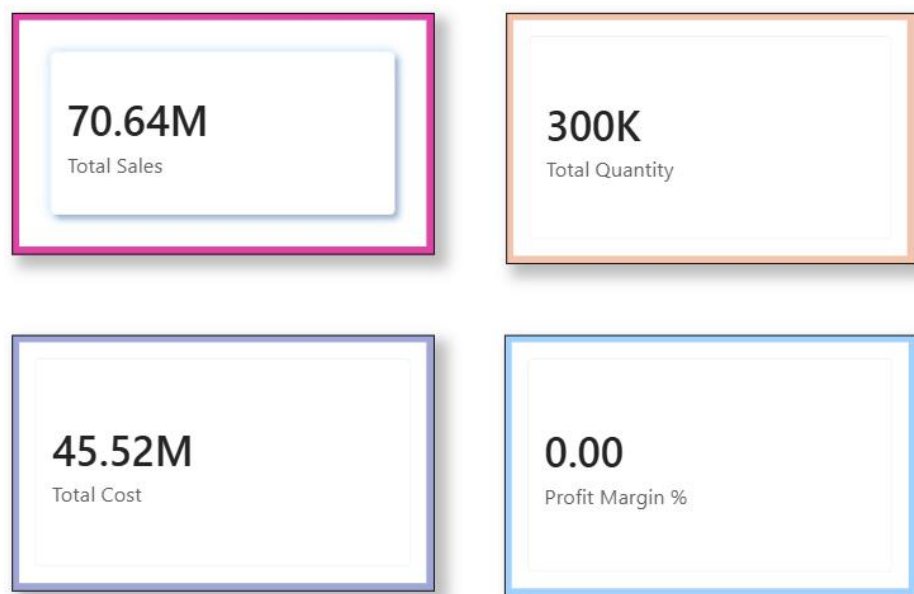
- **Total Quantity** – Calculates the total number of units sold across all products and sales channels.

- **Total Sales** – Calculates the overall sales revenue generated from the transactions in the dataset.
- **Total Cost** – Calculates the total cost associated with the products sold.
- **Profit Margin** – Measures the profitability of the business by expressing profit in relation to sales.

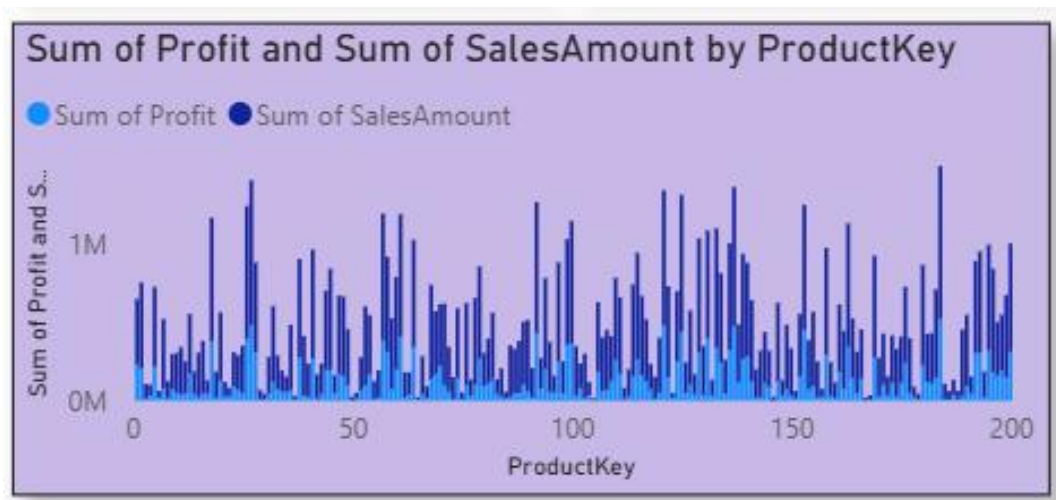
DASHBOARD COMPONENTS:

The dashboard consists of a combination of **KPI cards, charts, and interactive slicers** designed to provide a comprehensive view of global sales and financial performance. Each component presents a specific aspect of the dataset, such as sales, profit, products, order priority, and sales channels. These visualizations work together to make the data easier to understand, compare, and analyse.

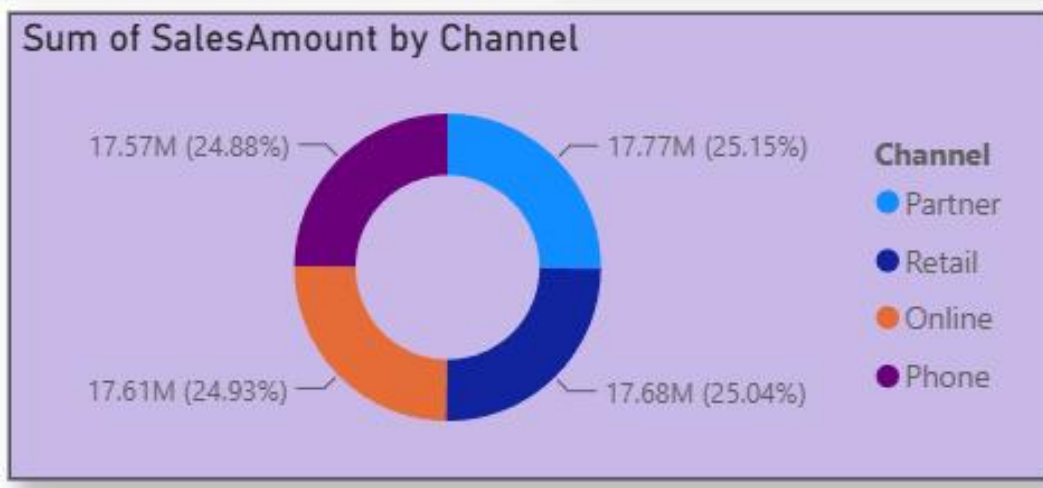
KPI Cards



- **Total Quantity – 300K:** Displays the total number of units sold across the dataset, providing an overview of overall sales volume.
- **Total Sales – 70.64M:** Represents the total revenue generated from all sales transactions.
- **Total Cost – 45.52M:** Shows the total cost associated with the products sold.
- **Profit Margin – 0.00:** Displays the calculated profit margin percentage, which is used to assess profitability in relation to sales.



2.SUM OF PROFIT AND SALES AMOUNT BY PRODUCT KEYA **column chart** is used to compare the **sum of Profit and Sum of Sales Amount** across different Product Keys. The Product Key is shown on the X-axis, while the corresponding financial values are represented on the Y-axis.



- Helps compare sales revenue and profit across individual products.
- Shows the variation in financial performance among different Product Keys.
- Helps identify products with relatively higher sales and profit values.
- Provides a product-level view of the company's financial performance.

3. SUM OF SALES AMOUNT BY CHANNEL

A doughnut chart is used to show the distribution of total sales across different sales channels: Partner, Retail, Online, and Phone.

- **Partner:** 17.77M (25.15%)
- **Retail:** 17.68M (25.04%)
- **Online:** 17.61M (24.93%)
- **Phone:** 17.57M (24.88%)

The sales contribution of the four channels is closely distributed, with each channel contributing approximately one-fourth of total sales.

3.SUM OF PROFIT BY ORDER PRIORITY

A horizontal bar chart is used to compare the total profit generated across different order priority levels: Low, Medium, and High.

- **Low priority** orders show the highest total profit, at approximately **11M**.
- **Medium priority** orders generate approximately **9M** in

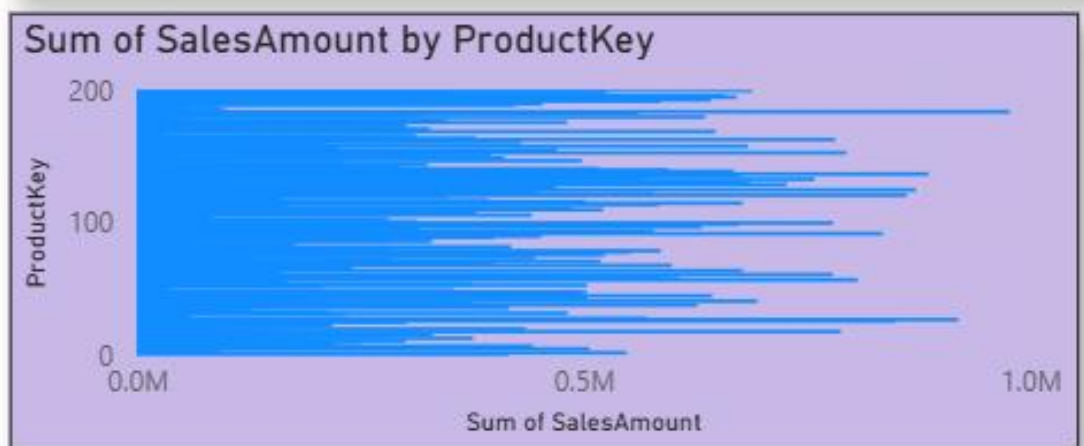


profit.

- **High priority** orders contribute approximately **4M** in profit.
- The chart clearly shows differences in profitability across the three priority categories.

4. SUM OF SALES AMOUNT BY PRODUCT KEY

A horizontal bar chart is used to display the total sales amount associated with different Product Keys.



- Provides a detailed view of **product-level sales performance**.
- Helps compare the sales contribution of individual products.
- Longer bars represent Product Keys with relatively higher sales amounts.

- The visualization highlights the **variation in sales across products**.
- It can help identify products that contribute more significantly to overall sales.

Purpose: To examine **product-wise sales performance** and understand how individual products contribute to total revenue.

5.SLICERS USED

The dashboard uses **three range slicers** to make the analysis interactive. These slicers allow users to filter the dashboard based on specific financial and sales-related values

The image displays three vertical range slicers, each with a title, two input boxes, and a slider. The top slicer is titled 'Profit' and has input boxes for '-177.02' and '2,852.08'. The middle slicer is titled 'Quantity' and has input boxes for '1' and '11'. The bottom slicer is titled 'UnitPrice' and has input boxes for '7.92' and '667.65'. Each slicer features a horizontal line with a circular handle at each end, indicating a range selection mechanism.

- **Profit Slicer:** Allows users to select a specific **profit range**, from approximately **-177.02 to 2,852.08**. This helps analyse records based on different profit levels.
- **Quantity Slicer:** Allows users to filter the data based on the **quantity sold**, ranging from **1 to 11**. This helps examine sales performance according to the number of units sold.
- **Unit Price Slicer:** Allows users to select a specific **unit price range**, from approximately **7.92 to 667.65**. This helps analyse products based on their selling price.

CONCLUSION:

The Global Sales and Financial Performance Dashboard demonstrates how Power BI can transform sales and financial data into meaningful visual insights. The dashboard provides a clear view of sales, costs, quantity, profit, product performance, order priority, and sales channels through interactive visualizations and slicers. Overall, the project shows how Power BI can support effective business analysis and data-driven decision-making.